MILITARY SPECIFICATION

CONTROLLER, ALTITUDE HOLD

This Military Specification has been approved by the Bureau of Naval Weapons Department of the Navy.

1. SCOPE

1.1 Scope. This specification establishes the general design and performance requirements of an automatic altitude hold controller for use in the MQM-36A target. The altitude hold controller shall be capable of operating in conjunction with Radio-Receiver-Decoder-Autopilot Type AN/ARW-79.

1.2 Classification. The altitude hold controller shall consist of the following:

Unit	Applicable Paragraph
Altitude Hold Controller	3.4.1

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of contract form a part of this specification to the extent specified herein:

Specifications

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring: Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Radio Frequency Interference Requirements

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MIL-S-8516 Sealing Compound, Synthetic Rubber, Electronic Connectors and Electric Systems

MIL-P-17555 Preparation for Delivery of Electronic, Electrical, and Electro-Mechanical Equipment and Associated Repair Parts

MIL-T-18303 Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for

2.2 Precedence of Documents. When the requirements of the contracts, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract. The contract shall have precedence over any specification.

(2) This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

2.3 Availability of Documents.

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

(2) Copies of this Specification may be obtained upon application to the procuring activity.

3. REQUIREMENTS

3.1 Parts and Materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and material as approved by Specification MIL-E-5400 shall be given first consideration.

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(2) When the characteristics of standard MIL parts and materials are such their use will not fulfill the requirements because of size, weight, performance or other reasons, material and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specification.

(3) When a nonstandard part is used, the meeting of general material requirements such as nonflammable, fungus resistant, nontoxic, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-E-5422 shall be of prime consideration.

3.1.1 Nonstandard Parts and Materials. Commercial grade parts and materials suitable for the fulfillment of the design objectives may be utilized, provided that strength, serviceability, reliability, and parts shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices. Diodes shall be chosen and applied, outlined in Specification MIL-E-4682.

3.2 Design and Construction. No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order.

3.2.1 General. The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship.

3.2.2 Total Weight. The total weight of the equipment shall be a minimum consistent with good design and shall not exceed seven and one-half pounds including connectors, mounts, and cables.

3.2.3 Operating Life.

3.2.3.1 Operational Stability. The equipment shall operate with optimum performance for ten hours, continuously or intermittently.

3.2.3.2 Total Operating Life. The equipment shall have a minimum total operating life of 100 hours with reasonable servicing and replacement of parts.

3.2.4 Cabling and Connections.

3.2.4.1 Cables and Connectors. The equipment shall provide for the use of cables and connectors in accordance with specification MIL-E-5400.

3.2.4.2 Interconnection Cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the appli-

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cable requirements of specification MIL-W-5088 except that in lieu of the requirements of paragraph 3.9.3, each cable assembly shall be marked with its cable number. The external wiring shall be unshielded.

3.2.5 Interchangeability. The equipment shall meet the interchangeability requirements as defined in specification MIL-E-5400.

3.2.6 Interference Control. The altitude hold equipment shall meet the requirements of MIL-I-6181.

3.2.7 Nameplates and Nomenclature. Nameplate material shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Vibration	None
Temperature	Room ambient ($30^{\circ} \pm 20^{\circ}C$)
Altitude	Normal ground
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	27.5 ± 0.5 v.d.c.

3.2.9 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.9.1 Vibration. The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve II of specification MIL-E-5400.

3.2.9.2 Temperature. The equipment shall operate satisfactorily when subjected to the temperature requirements of Table II of specification MIL-E-5400 with $-35^{\circ}C$ as minimum temperature.

3.2.9.3 Altitude. The equipment shall operate satisfactorily when subjected to the altitude requirements of Table II of specification MIL-E-5400, up to 15,000 feet.

3.2.9.4 Shock. The equipment shall operate satisfactorily when sub-

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jected to the shock requirements of paragraph 3.2.21.6.1 of specification MIL-E-5400.

3.2.10 Waterproofing. The equipment shall be capable of immersion in salt water to a depth of three feet for a period of sixty minutes after descent from maximum operating altitude without damage to the internal parts or the admission of salt water. Electrical connectors shall be waterproofed by filling the back shell cavity with a potting compound in accordance with MIL-S-8561.

3.2.11 Input Power Requirements. The equipment shall be powered from the existing 28 volt battery in the target and shall operate satisfactorily with voltages between 22.5 and 30 volts. The current drain of the equipment shall not exceed 1.5 amperes.

3.3 Performance.

3.3.1 General. Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions.

3.3.2 Compatibility. The equipment shall be capable of operation in conjunction with the existing AN/ARW-79 Remote Flight Control System, consisting of a proportional control receiver-decoder autopilot AN/ARW-79, a type K-7 vertical gyro, and two type D-9 servos.

3.3.3 Operation. Under normal operating conditions, a KD-2R-5 target shall maintain a stable flight altitude when it is trimmed and then the Altitude Hold engaged. The equipment shall maintain a constant pressure altitude of plus or minus 5% of reference altitude or plus or minus 100 feet, whichever is greater, for any 15 minutes of flight between 200 and 15,000 feet reference altitude with 50 to 100 feet momentary loss in turns.

3.3.4 Remote Control. The altitude hold controller shall be engaged and disengaged at anytime during flight by remote control via the existing command radio receiver. The altitude hold controller shall be automatically disengaged upon termination of flight.

3.3.5 Altitude Hold Command. The altitude hold controller shall be alternately engaged and disengaged in response to successive commands received by the spare channel of the radio receiver. Pin "B" of Power Receptacle J1 is connected to ground through a relay contact in the radio receiver when the spare channel is operated.

3.3.6 Recovery Command. The altitude hold controller shall be automatically disengaged when a recovery command is received by the radio receiver. Pin "D" of Power Receptacle J1 in the radio receiver is connected

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to ground through a recovery switch in the junction box when recovery of the target is commanded.

3.3.7 Altitude Hold Controller Engaged. An air chamber contained in the controller shall be closed to the atmosphere by means of a solenoid valve when a command to engage the system is received by the spare channel of the radio receiver. The altitude control signal shall be proportional to the differential pressure caused by a change in altitude from the referenced altitude pressure existing in the trapped-air chamber. When the target aircraft deviates from the altitude to be maintained, an electrical wiper shall be moved along a potentiometer to provide a change in resistance proportional to the pressure change.

3.3.8 Altitude Hold Controller Disengaged. The trapped-air chamber contained in the controller shall be opened to the atmosphere by means of the solenoid valve when a command to disengage the system is received by the spare channel of the radio receiver. The potentiometer wiper shall return to the electrical center of the potentiometer and the AN/ARW-79 radio receiver-decoder autopilot shall function in the normal manner. The target aircraft may then be commanded in pitch to a new altitude.

3.3.9 Indicator. The altitude hold controller shall be equipped with a mechanical visual indicator for the "off" or "on" condition. A white signal shall indicate "off" and a red signal shall indicate "on".

3.4 Detail Design Requirements.

3.4.1 Altitude Hold Controller. The altitude hold controller shall meet the following requirements:

3.4.1.1 Function. The altitude hold controller shall maintain the MQM-36A target at a constant pressure altitude as specified in paragraph 3.3.3 when engaged by remote radio control. This action shall be accomplished by actuation of a differential pressure transducer having a potentiometer type pick-off serving to control the elevator servo of the target.

3.4.1.2 Form Factor. The altitude hold controller shall not exceed the following dimensions and shall fit in MQM-36 target as shown in installation drawings (Figures 1 and 2):

height	10.50 inches
width	9.0 inches
depth	6.70 inches.

3.4.1.3 Weight. The total weight of the altitude hold controller shall not exceed seven and one half pounds including connectors, mounts, and cables.

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3.4.1.4 Components of Unit. The altitude controller shall be comprised of the following components and parts:

3.4.1.4.1 Control Indicator. A control-indicator shall be provided which engages or disengages the altitude controller as required in 3.3.7 and 3.3.8. The unit shall contain a rotary switch, which performs the engage/disengage switching function and at the same time provides a mechanical visual indication in accordance with paragraph 3.3.9; three standard resistors rated at 5% tolerance and $\frac{1}{2}$ watt power dissipation; and a standard diode for arc suppression.

3.4.1.4.2 Mounting Platform. A mounting platform shall be supplied upon which the major components are mounted.

3.4.1.4.3 Differential Pressure Transducer. A differential pressure transducer for an external trapped-air volume of 300 cc and containing a potentiometer-type pick-off having a total resistance of 1120 ohms $\pm$ 5 percent shall be provided. The transducer shall be equipped with a latching valve which closes and opens the (external) air chamber as the altitude hold controller is engaged and disengaged in accordance with paragraphs 3.3.7 and 3.3.8 of this specification. The altitude control signal supplied by the transducer shall be linear and proportional to the differential pressure caused by a change in altitude from the reference altitude pressure existing in the external air chamber.

3.4.1.4.4 Temperature Compensated Air Chamber. An air chamber shall be provided having the capability of maintaining at an essentially constant pressure the air trapped within the chamber over a temperature range of -20°C to +55°C.

3.4.1.4.5 Shut-Off Valve. A shut-off with provisions for accepting a lanyard pin whose withdrawal closes the valve shall be furnished. The valve shall be installed in line with the static inlet pressure port of the controller and shall serve the purpose of preventing water from entering the controller through the air lines upon immersion in water.

3.4.1.4.6 Air Pressure Fittings and Tubing. Air pressure fittings and tubing for application at atmospheric pressures from sea level to 15,000 ft. altitude, shall be provided for interconnection between the air chamber and the differential pressure transducer.

3.4.1.4.7 Cables. The altitude controller shall be equipped with two cables. One cable is to interconnect the differential pressure transducer to the control-indicator, and the second provides for interconnecting the altitude controller to the MQM-36A Control System.

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3.4.2 Auxiliary Equipment. The following auxiliary equipment shall be provided with the altitude controller:

3.4.2.1 Mounting Hardware. Mounting hardware shall be provided to permit installation of the altitude controller in the target.

3.4.2.2 Static Lines and Ports. Plastic tubing and ports shall be provided to apply atmospheric pressure to the static inlet port of the altitude controller.is required.

3.5 Preproduction Samples. Unless otherwise specified in the ntract or order, preproduction units of the altitude hold controller all be manufactured using the methods and procedures proposed for the oduction. The samples will be tested as specified in Section 4 herein r the purpose of determining that, prior to starting production, the ntractor's production methods are capable of yielding items that comply th the technical requirements of the contract.

QUALITY ASSURANCE PROVISIONS

4.1 Classification of Inspection. Inspection of the altitude ld controller shall be classified as follows:

a. Preproduction Inspection

b. Acceptance Inspection.

4.1.1 Responsibility for Inspection. Unless otherwise specified the contract or purchase order, the supplier is responsible for the rformance of all inspection requirements as specified herein. Except otherwise specified, the supplier may utilize his own facilities or y commercial laboratory acceptable to the Government. The Government serves the right to perform any of the inspections set forth in the ecification where such inspections are deemed necessary to assure sup- ies and services conform to prescribed requirements.

4.2 Preproduction Tests. Preproduction tests shall be made on e or more equipments representative of the production equipments to be pplied under the contract. Preproduction tests shall consist of the llowing: (1) Contractor's Demonstration Tests; (2) Service Approval sts.

4.2.1 Contractor's Demonstration Tests. Contractor's demonstra- on tests shall be accomplished under the responsibility of the contrac- r and shall be conducted in accordance with the approved test procedure

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Contractors not having adequate facilities to conduct all required tests shall obtain the service of a commercial testing laboratory satisfactory to the procuring activity.

4.2.2 Service Approval Tests. At the completion of the contractor's demonstration test, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.3 Scope of Tests. Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with specification MIL-T-5422.

4.2.4 Preproduction Approval. Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be subject to the approval of the Government Inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests.

4.3.1 Individual Tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workman-

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ship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test.

4.3.1.1 Examination of Product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.2 Sampling Tests. Sampling tests shall be conducted at the contractor's plant on complete equipments which have successfully passed the individual tests. The equipments shall be selected by the Government Inspector. Tests shall be conducted on a quantity as approved by the procuring activity, shall be more extensive than the individual tests, and may include any of the tests of paragraph 4.2.1 deemed necessary by the Government Inspector to determine that the equipment offered for acceptance continues to be equivalent in performance and construction to the approved preproduction model.

4.4 Presubmission Testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.5 Rejection and Retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.6 Equipment Changes. After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 General. The equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in

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accordance with specification MIL-P-17555. The preservation shall be Method II of MIL-P-116, for level "A" packaging.

6. NOTES

6.1 Use. The equipment specified herein will be installed in an MQM-36A target aircraft and will maintain the target at a constant pressure altitude when the altitude hold system is engaged.

6.2 Test Values. Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 v.d.c. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

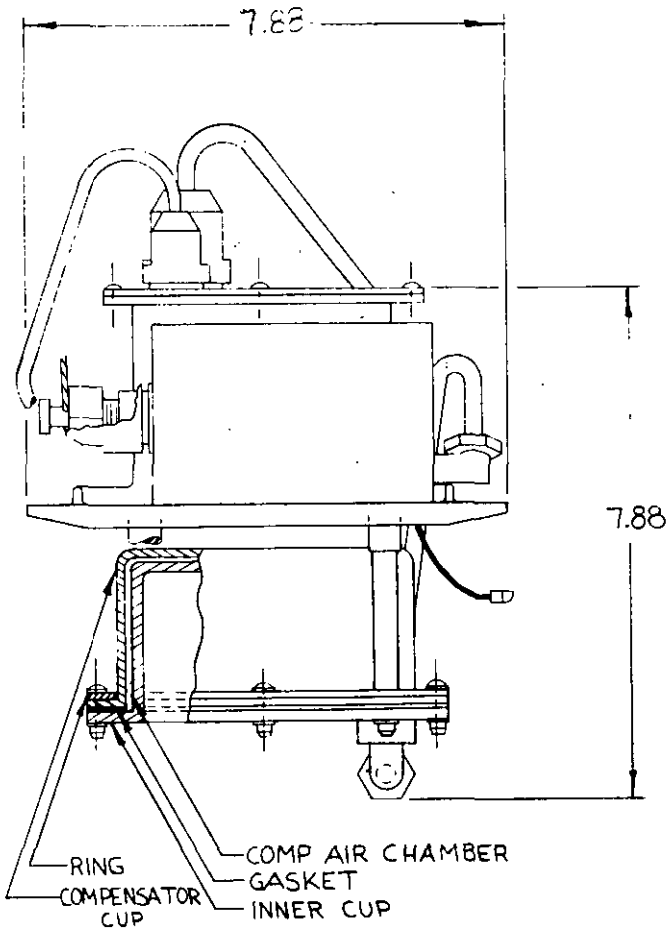
6.3 Ordering Data. Contract procurement documents should specify the following:

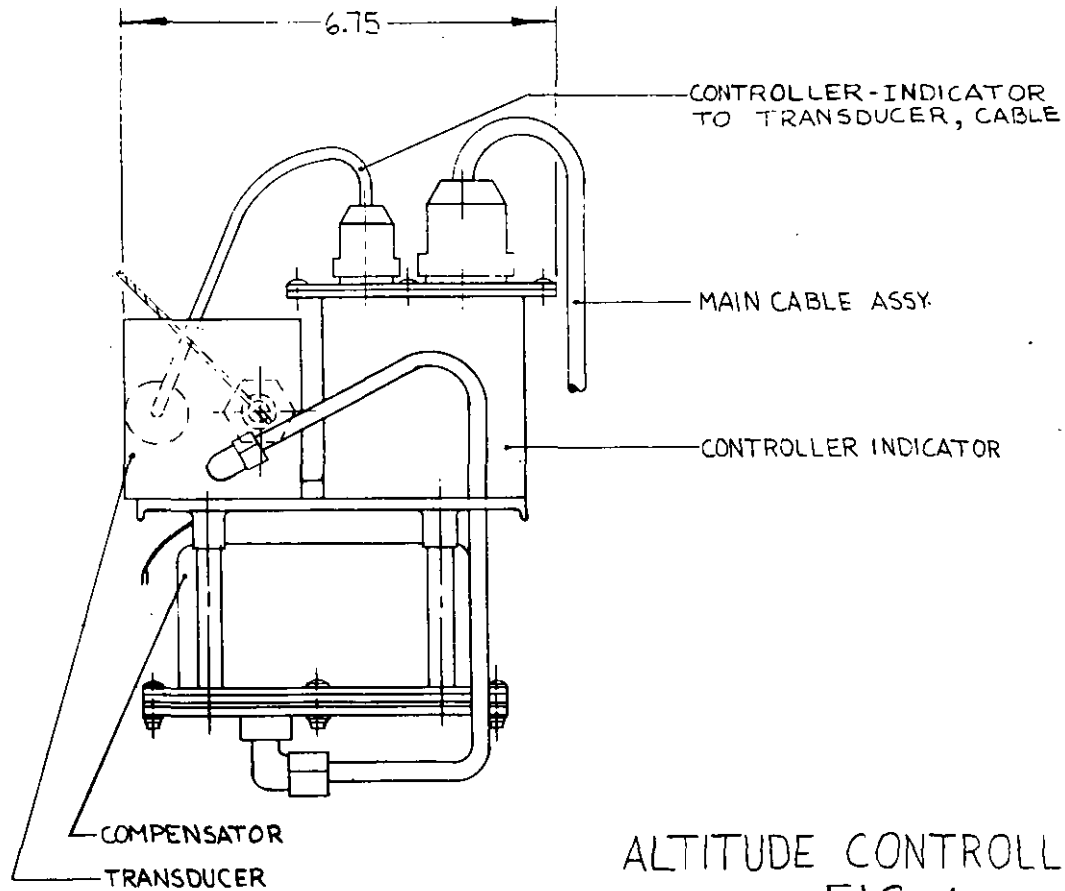
(a) Title, number and date of this specification.

(b) Selection of applicable level of packaging and packing required (see 5.1).

(c) Data required for operational use and for maintenance and procurement purposes shall be obtained in accordance with paragraph 6.2 of Military Specification MIL-D-18300C (WEP) dated 15 Dec 1961.

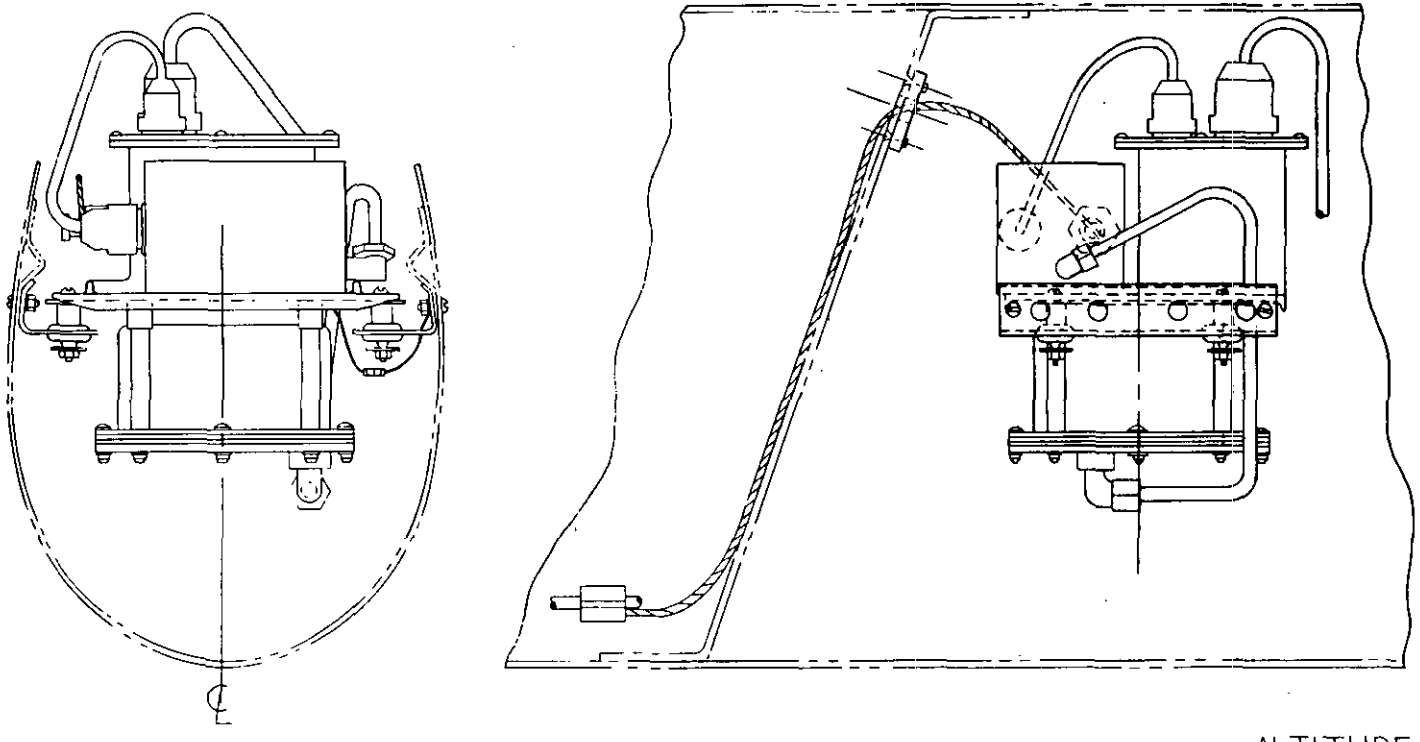
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ALTITUDE CONTROLLER
FIG. 1

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ALTITUDE HOLD CONTROLLER & AUXILIARY EQUIPMENT (AS INSTALLED)
FIG. 2

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